WILEY

British Ecological Society

Canopy Seed Bank Dynamics and Optimum Fire Regime for the Highly Serotinous Shrub, Banksia Hookeriana

Author(s): N. J. Enright, Byron B. Lamont and Ralf Marsula

Source: *Journal of Ecology*, Vol. 84, No. 1 (Feb., 1996), pp. 9-17
Published by: British Ecological Society
Stable URL: http://www.jstor.org/stable/2261695
Accessed: 07-05-2016 01:46 UTC

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at
http://about.jstor.org/terms

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Wiley, British Ecological Society are collaborating with JSTOR to digitize, preserve and extend access to *Journal of Ecology*

This content downloaded from 209.175.73.10 on Sat, 07 May 2016 01:46:40 UTC
All use subject to http://about.jstor.org/terms

Journal of Ecology 1996, **84**, 9–17

Canopy seed bank dynamics and optimum fire regime for the highly serotinous shrub, *Banksia hookeriana*

N.J. ENRIGHT,*† BYRON B. LAMONT† and RALF MARSULA‡

**Department of Geography and Environmental Studies, University of Melbourne, Parkville, 3052, Australia, †School of Environmental Biology, Curtin University of Technology, GPO Box U1987 Perth, 6001, Australia and ‡Sektion Ökosystemanalyse, UFZ-Umweltforschungszentrum, Leipzig-Halle GmbH, Permoserstr. 15, PF 2, 0-7050 Leipzig, Germany*

Summary

1 *Banksia hookeriana* is a serotinous nonsprouting shrub restricted to the fire-prone northern sandplain scrub-heaths of Western Australia. Evidence of the natural fire frequency to which the species is best adapted was sought through analysis of its seed demography using a computer model of canopy seed bank dynamics based on field data.

2 Plants began flowering at 3–4 years of age but storage of viable seeds in the canopy was rare for plants < 5 years. Infructescences (cones) contained a mean of 12 follicles, holding two seeds of which 62% were estimated to be viable.

3 Cone (and seed) production increased with plant age to 15 years, but then fluctuated widely around a mean of 17 cones per year. By this time the average plant had an estimated canopy seed bank of 700 viable seeds comprising seeds 1–10 years old.

4 Seeds were lost from the canopy seed bank through spontaneous follicle rupture (i.e. in the absence of fire), granivory by insect larvae, and decay. Most follicles remained closed for at least 5 years and up to 50% of follicles were still closed after 10 years. Insect granivores destroyed 18% of seeds in 1-year-old cones, and continued to destroy seeds in older cones at a rate of 2–3% per year.

5 The seed bank model predicted continued seed accumulation in the canopy to at least age 25 years (1400 seeds), but by this time more viable seeds ($\approx$ 2200) had been lost to spontaneous opening, granivory and decay than were then stored on the plant.

6 Shannon–Weiner diversity (H′) of seed ages in the canopy seed bank increased with plant age, while evenness (V′) decreased at first (to plant age 17 years) and then increased.

7 On a time-scale of many generations, the likelihood of successful recruitment of individuals in the next generation (on average), occurs for a fire frequency of 15–18 years. At this frequency resource wastage in this low-nutrient environment would be minimized.

8 Recent fires have recurred within 7–11 years and indicate a human impact on the natural fire regime which may ultimately threaten the species.

Keywords: *Banksia*, fire frequency, seed bank model, seed demography, serotiny

Journal of Ecology (1996) **84**, 9–17

Introduction

Serotiny refers to the retention of seeds in closed fruits within the plant canopy for more than one year, and can be contrasted with two alternative fates of seeds: storage in a soil seed bank, or release of short-lived seeds upon maturity (no seed storage) (Lamont *et al.* 1991a). Seed release in serotinous species is normally triggered by fire and seeds are short-lived after release, germinating during the first favourable period (Cowling & Lamont 1987). Species can range from weakly serotinous, where most seeds are released within a few years of production, to strongly serotinous, where seeds are held in the canopy for many years. The

Correspondence: Dr Neal Enright, Department of Geography and Environmental Studies, University of Melbourne, Parkville, 3052, Australia, Fax: 61 39344 4972, E-mail: Enright@ariel.ucs.unimelb.edu.au

© 1996 British Ecological Society

This content downloaded from 209.175.73.10 on Sat, 07 May 2016 01:46:40 UTC
All use subject to http://about.jstor.org/terms

ecological significance of serotiny seems clear: where conditions for seedling recruitment are best immediately following fire (i.e. competition for light, moisture and nutrients are probably at their minimum), cued release of canopy-stored seeds after fire maximizes the age, and therefore canopy seed store, of the population by the time of the next fire. Any seedlings that establish later during the interfire period have a low probability of survival due to competition with their parents and other established plants (Cowling & Lamont 1987; Lamont & Barker 1988). They will be much younger and, on average, will have a smaller canopy seed bank when fire recurs and so are less fit (*sensu* Stearns 1992).

The dynamics of seed production, storage and loss in relation to plant age determines the range of fire frequencies for which population maintenance is possible. If fire occurs either before the onset of reproductive maturity, or before a sufficient canopy seed bank has accumulated, then a population is threatened with extinction.

In the absence of fire some seeds are released by the spontaneous opening of old fruits, but the likelihood of plant recruitment and survival to reproductive age is extremely low. Seed release is hastened by the death of stem tissue due to drought stress and plant senescence (Witkowski *et al.* 1991). This means that the canopy seed bank may be completely lost where plant death occurs before the next fire, a fate not shared by most soil seed bank species where seeds may survive for many years after the death of parent plants (Baker 1989). Because dry plant biomass is more readily consumed by fire, any seeds held in the canopy of dead plants at the time of fire are also likely to perish (Lamont & Barker 1988).

Banksia hookeriana Meissn. (Proteaceae) is a highly serotinous, fire-killed (nonsprouting) woody shrub to 2.5 m high which is restricted to the northern sandplain scrub-heaths of south-western Australia about 250–330 km north of Perth (George 1981; Taylor & Hopper 1988). Seeds are held in a state of enforced dormancy within woody follicles on large infructescences (cones) in the crown. While new inflorescences may be damaged and removed from plants by birds, mature cones normally remain attached to the plant throughout its life. Some seeds abort during development, while others are either released gradually or (especially after fire) are consumed by insects, lose viability and decay, or remain viable. Once released seeds germinate with the onset of winter rains or perish as there is no persistent soil seed bank (Enright & Lamont 1989).

The geographical extent (80 × 30 km) of the low nutrient, noncalcareous, sandplain defines the distribution of this species, which is further limited by its specific preference for the deeper sands of low dune crests (Enright & Lamont 1989; Lamont *et al.* 1989). This means that the species exists as a large number of discrete populations of varying sizes (< 100 to > 10 000 individuals) separated by distances of 0.2 to > 1 km. The largest populations are located near Mt Adams in the northern part of the species range. Land clearance for agriculture, mining and mineral exploration, and increased incidence of anthropogenic ignition, make habitat loss and increased fire frequency potential threats to the long term survival of this species. In addition, its attractive orange inflorescences are harvested in large numbers for the commercial wildflower trade, so reducing seed production in exploited populations (Witkowski *et al.* 1994).

This paper describes seed production and availability, and a computer model of seed bank dynamics in *Banksia hookeriana* in relation to plant age (= time since last fire) for sites near Mt Adams. The following questions were posed:

1 What is the pattern of seed production with plant age, and what factors influence the rate of seed accumulation in the crown (e.g. level of serotiny, granivory, seed abortion and longevity)?
2 Does the analysis of seed demography reveal the optimum fire frequency to which the species is adapted, and the range of fire frequencies compatible with population survival?

STUDY AREA

The study was located in the Yardanogo Nature Reserve about 330 km north of Perth (29°25′S, 114°58′E). Mean annual rainfall at the two closest climate stations is 513 mm (Eneabba; 55 km to the south) and 471 mm (Dongara; 25 km to the north), respectively. There is a pronounced winter maximum with almost no rain (median 20 mm) in the five summer months from December to April. Temperature regimes are similar for both climate stations: mean maximum temperature in the hottest month (January) is 35.8 °C, mean minimum temperature in the coolest month (July) is 9.2 °C. Soils vary from acidic, leached sands on low dunes to sandy clays in intervening depressions (Hnatiuk & Hopkins 1981). The vegetation of the low dunes is a diverse scrub-heath usually dominated by two *Banksia* species, *Banksia hookeriana* and *Banksia attenuata* R. Br., although up to six banksias may co-occur (Beard 1984; Enright & Lamont 1989).

Materials and methods

Seven sample plots, each 400 m^2 in area and ranging in age since last fire from 1 to 3 and 14–16 years, were established in 1986 and monitored for 6–8 years. They were on the crests and slopes of low, sandy dunes in scrub-heath along the edge of a 2-km section of Mt Adams Road within the Yardanogo Nature Reserve. Sites covering a broader range of *Banksia hookeriana* stand ages could not be located.

© 1996 British Ecological Society,

This content downloaded from 209.175.73.10 on Sat, 07 May 2016 01:46:40 UTC
All use subject to http://about.jstor.org/terms

For the five plots where plant age was 1–3 years at the start of the study, and when fire occurred subsequently during the study period (wildfire killed all plants in four of these five plots in 1991), age at time of last fire was estimated for 10 plants by the stem node ageing method described by Lamont (1985), or was known precisely. Seed production was estimated by counting the total number of cones on a random sample of 10 plants and the number of follicles on each of 10 randomly chosen cones per plant. The onset and level of flower head (inflorescence) and cone (infructescence) production were recorded within five replicate 25-m^2 subplots per plot for each year until 1991 (four plots) or 1993 (one plot).

For the two plots where plant age was 14–16 years in 1986, the total number of plants and total number of cones per plant was determined. For a random sample of 10 plants per plot, the height and crown diameter, and number of new cones was enumerated each year there-after until 1991 (when wildfire killed all plants in one plot) or 1993. Plots were searched each year for evidence of seedling establishment from spontaneous release of seeds in the absence of fire.

Complete cone crops were harvested from 10 randomly chosen 16-year-old plants at the start of the study in 1986. Cones were allocated to annual age classes using the stem node ageing method so that the entire history of reproduction could be determined for each plant. Degree of serotiny was based on the proportion of follicles per cone age class which had opened in the absence of fire (Cowling & Lamont 1985a). Cones were burned and subjected to wet-dry cycles to induce follicle rupture and facilitate seed removal (Cowling & Lamont 1985b). The numbers of follicles, and proportions of firm, aborted, eaten and decayed seeds were determined per plant by cone age. Viability of firm seeds was assessed by germinating batches of 25 seeds per cone age class from each of five plants (replicates) under moist conditions at 15 °C (Cowling *et al.* 1987).

STATISTICAL ANALYSES

Trends between seed parameters, plant age and cone age were analysed using regression models (SYSTAT 1992). Where data were not normally distributed, logarithmic (ln) and arcsine transformations were employed to satisfy assumptions of the regression models used. All error terms listed are standard errors.

A canopy seed bank dynamics model, written in programming language C++, was used to estimate the pattern of viable seed retention on plants and fate of seeds in relation to age. The model used regression equations and other statistics relating patterns of seed accumulation and loss to increasing plant and cone age. Shannon–Weiner diversity (H′) and evenness (V′) were calculated for the distribution of viable seed ages by plant age.

Results

CONE PRODUCTION

A few plants produced inflorescences at 3–4 years of age, but these were all sterile (age 3: mean inflorescences per plant = 0.12 ± 0.06; age 4: mean = 0.92 ± 0.23). Plants first produced fertile cones at an age of 5 years. Cone production increased until plants were about 15 years old, after which annual production fluctuated widely about an average of 17 cones per year (Fig. 1a; Table 1). Both within and between year variability in cone production per plant increased with plant age.

The total number of cones held in the crown increased continuously with plant age and although the standard error also increased with age, the mean pattern of accumulation showed a consistent trend (Fig. 1b, Table 1).

SEED PRODUCTION

Cones of *Banksia hookeriana* contained up to 20 woody follicles, each of which held two seeds. The maximum possible number of seeds on a plant can

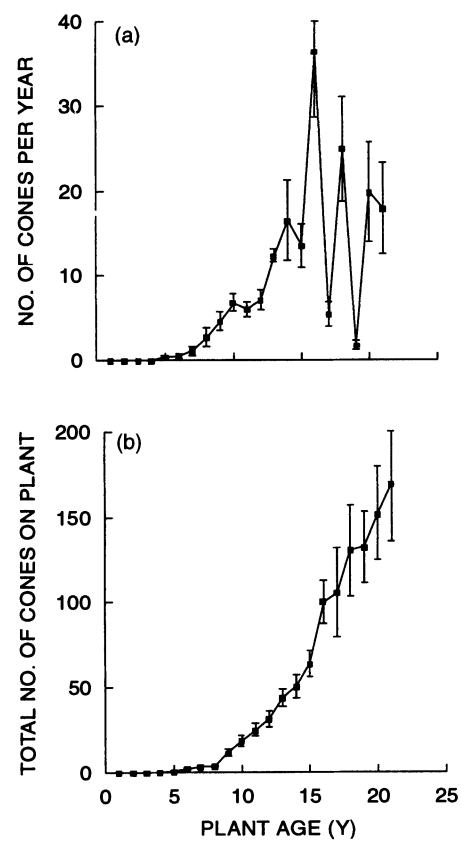


Fig. 1 (a) The number of cones produced per year in relation to plant age, and (b) the total number of cones held in the crown in relation to plant age for *Banksia hookeriana* near Mt Adams, Western Australia. Note: standard errors are calculated for the 10 plants which were aged and harvested at the start of the study in 1986, and subsequently for total cones on random sets of 10 plants each year to 1993.

© 1996 British Ecological Society,

This content downloaded from 209.175.73.10 on Sat, 07 May 2016 01:46:40 UTC
All use subject to http://about.jstor.org/terms

Table 1 Equations describing the pattern of seed accumulation and loss in relation to plant and cone age (as numbered in Fig. 3) for *Banksia hookeriana* near Mt Adams, Western Australia. Parameters related to cone age were derived from data for all cones on 10 16-year-old plants

Regression model	d.f.	r	F	P
eqn 1. New cones per year by plant age ln (# cones) = 3.61 (ln (plant age)) – 0.14 (plant age) –5.12	2,123	0.81	118.3	< 0.0001
eqn 2. Potential seeds per plant (a) Potential seeds = 12.08(2)(total cones on plant) (b) Potential seeds = 23.75(total cones on plant) –1.11	1,122	0.89	465.6	< 0.0001
eqn 3. Open follicles (proportion of total per cone age) ln (proportion follicles open) = 0.48(cone age) –5.27	1,94	0.85	236.6	< 0.0001
eqn 4. Potential firm seeds Firm seeds = 0.83(potential seeds)				
eqn 5. Insect-damaged seeds (proportion per cone age) Damaged seeds = 0.02(cone age) + 0.18	1,92	0.36	11.3	0.001
eqn 6. Decayed seeds (proportion per cone age) ln (decayed seeds) = 0.34(cone age) –5.95	1,91	0.66	70.1	< 0.001
eqn 7. Viable seeds Viable seeds = 0.744(firm seeds)				

therefore be described by the simple equation: number of potential seeds = (number of cones)(number of follicles)(2). On average, cones contained 12.08 ± 0.04 follicles, with no significant trend in relation to plant age (Fig. 2a). Although each follicle contained two seeds, only 83.0 ± 1.0% of seeds were firm (i.e. had developed embryos). Viability of firm seeds was variable between years (overall mean 74.4 ± 17.8%), but differences were not significant (Fig. 2b).

SEROTINY, GRANIVORY, DECAY AND SEED STORAGE

In the absence of fire nearly all follicles remained closed for the first 5 years. Thereafter the rate of spontaneous seed release increased linearly (Fig. 2c, Table 1). After 10–12 years the cones were heavily weathered and many of the seeds decomposed so that few viable seeds remained.

Since plants only began to set viable seed at an age of about 5 years, and little spontaneous follicle rupture occurred in cones less than 5 years old, there was almost no seed loss due to follicle rupture in plants up to 10 years old. Above this age, the percentage seed loss increased markedly: 5% of all follicles on plants were open by age 13 years, 10% by 16 years, and 20% by age 21 years. No seedlings which established in the sample plots as a result of interfire seed release survived for more than one year.

The decline in seed storage with cone age was also attributable to seed loss through granivory and decay (Table 1). Loss of seeds through fungal decay followed a similar pattern to that described for spontaneous seed release due to follicle rupture in the absence of fire. Decay was almost absent until cones were 6 years old, after which the proportion of seeds lost increased rapidly.

Granivory within closed follicles by the larvae of two weevils, *Cechides amoenus* (Curculionidae; Cleoninae) and *Eleale aulicodes* (Curculionidae; Cleridae), and a jewel beetle (Bupestridae; *Astraeus* sp.) was observed. Unlike follicle opening and decay, granivory was recorded for all cone ages (Table 1).

THE SEED BANK DYNAMICS MODEL

By integrating the information on average patterns of seed production, storage and loss in relation to plant and cone age, it was possible to model the pattern of storage of viable seeds on an average plant of any age, and to identify the quantities of seeds lost to incomplete seed development, senescence, interfire dispersal (spontaneous follicle opening), granivory and decay. The statistical relationships used in the model have been described above and are listed in Table 1, and a flow diagram of the model is shown in Fig. 3.

The model first estimated the number of new cones produced per year for each plant age up to 25 years using eqn 1 (Table 1). Number of cones per year was set to 17 once plant age reached 15 years (the average for years 15–21). Estimation was stopped at age 25 years since field data were only available for plants up to 21 years. The number of cones was summed for each possible plant age to produce an estimate of the total number of cones held in the crown at any plant age. The total number of seeds which could be held on the plant, if every follicle on every cone held two seeds, was then determined. Two estimation methods were available and gave almost identical results: one based on mean follicle number per cone for samples

© 1996 British Ecological Society,

This content downloaded from 209.175.73.10 on Sat, 07 May 2016 01:46:40 UTC
All use subject to http://about.jstor.org/terms

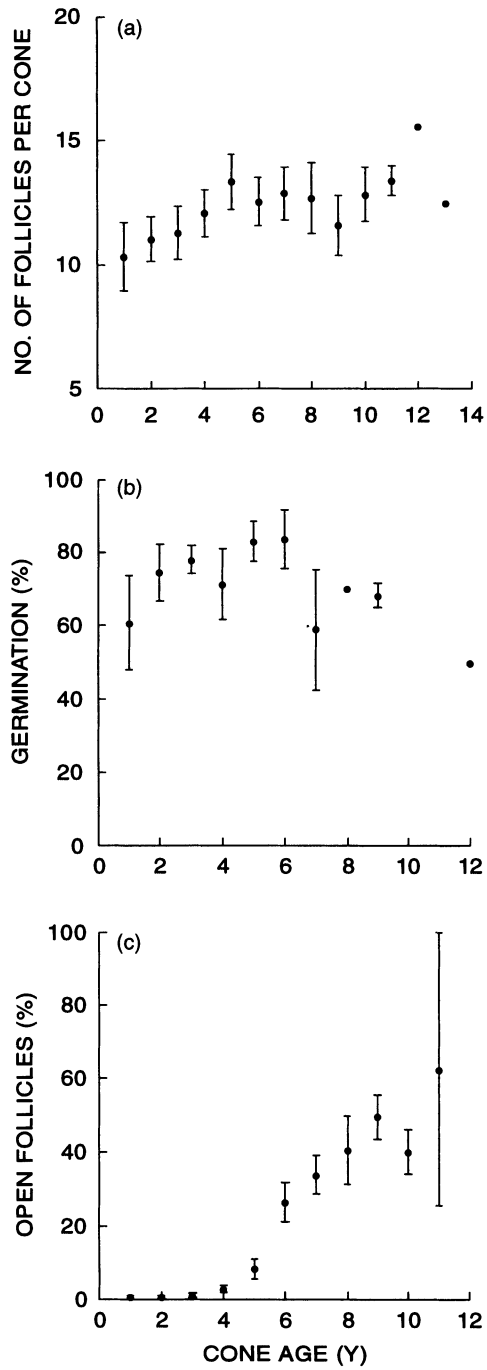


Fig. 2 (a) The number of follicles per cone (± SE) in relation to cone age, (b) percentage germination of firm seeds extracted from closed follicles on cones of different age, and (c) proportion of follicles per cone which opened spontaneously in the absence of fire in relation to cone age for *Banksia hookeriana* near Mt Adams, Western Australia. Note: error bars are not included for data points where sample size was less than 10 cones, or five replicate batches of seeds, per cone age class.

of 10 cones per plant (eqn 2a), and another based on complete follicle counts for all cones on a total of 123 plants (eqn 2b). The latter method of estimation was used in the model.

The proportion of seeds lost from cones due to spontaneous follicle rupture in the absence of fire was estimated for each cone age (eqn 3) and multiplied by the estimated number of seeds in that age class. A new total for remaining seeds by plant age was then determined. The total number of seeds remaining was multiplied by the proportion of firm seeds (eqn 4) to eliminate aborted seeds held in closed follicles from future calculations. The proportions of firm seeds lost to granivory and decay were estimated for each cone age class (eqns 5 and 6) and deducted from the seeds available within that age class. Finally, the estimated number of viable seeds remaining on a plant of given age was obtained by multiplying the total firm seeds after losses to granivory and decay by the germination rate for firm seeds.

Predicted total seeds produced per plant by age 25 years was 5800. Plants were likely, on average, to hold 100 viable seeds in closed follicles by the time they were 8 years old, 1000 by the time they were 17 years, and 1400 when they were 25 years (Fig. 4). The annual rate of viable seed accumulation in the crown peaked at 15 years. By the time plants were 25 years old, the combined impacts of follicle rupture in the absence of fire, and decay, resulted in little further net accumulation of viable seed in the plant canopy and more seeds were lost than were gained per year (Fig. 4). The overall canopy seed store represented a declining proportion of the total seeds produced, dropping from about one-half at age 8 years to one-quarter at age 25 years.

Shannon–Weiner diversity (H′) of seed ages represented in the canopy seed bank increased continuously with plant age following the onset of reproductive maturity; quickly at first as the number of seed ages represented increased on young plants ($H' = 0.66$ at 6 years, to $H' = 2.00$ at 15 years), and then more slowly as the number of seed ages represented on the plant stabilised ($H' = 2.28$ at 25 years). The continued increase in H′ after annual seed production had peaked (15 years) and the number of seed ages represented had stabilised (18 years), resulted from increasing evenness (V′) of seed abundance in each age class. Evenness was initially high ($V' = 0.95$ at 6 years), declined as each year's seed crop increased ($V' = 0.80$ at 17 years), then increased again as seed production stabilised ($V' = 0.88$ at 25 years).

Discussion

SEED DEMOGRAPHY

Flowering in *Banksia hookeriana* commenced at an age of 3–4 years, but few or no viable seeds were stored on plants < 5 years old, so that recurrent fire within this time would lead to loss of the species from the community. The time to first reproduction is similar to that described for a number of other serotinous, nonsprouting trees and shrubs in southern and eastern Australian scrub-heaths (Bradstock 1985;

© 1996 British Ecological Society,

This content downloaded from 209.175.73.10 on Sat, 07 May 2016 01:46:40 UTC
All use subject to http://about.jstor.org/terms

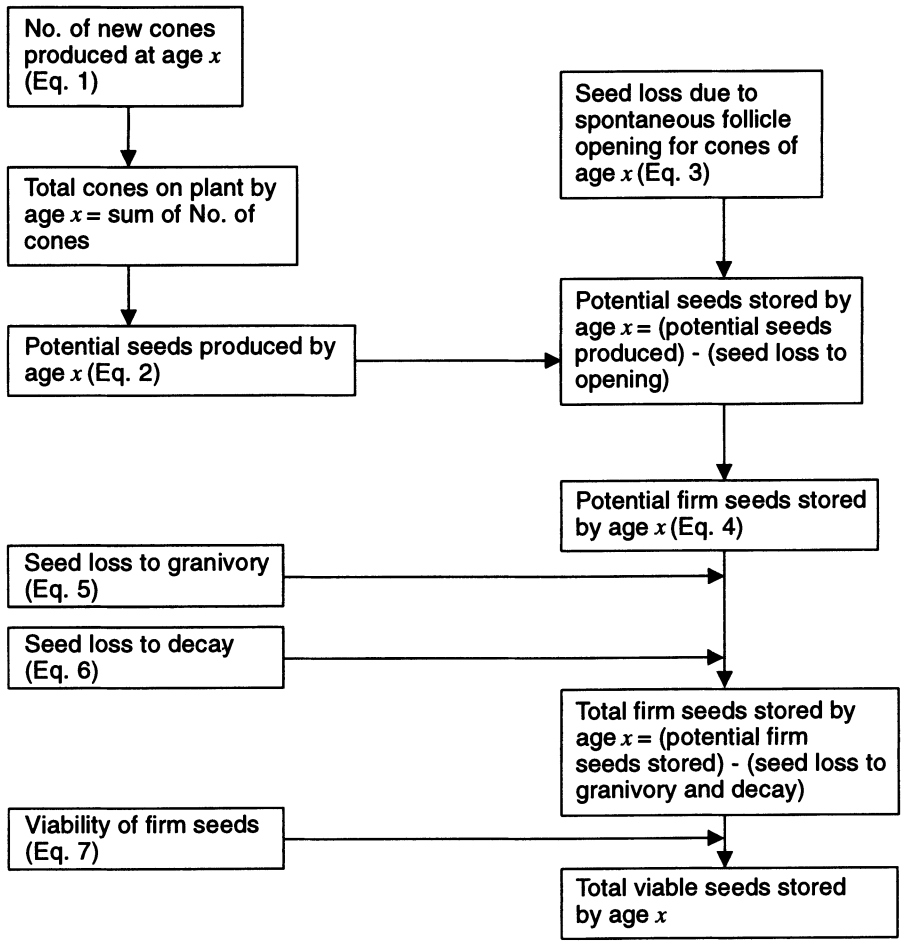


Fig. 3 Schematic diagram of the seed bank dynamics model for estimating the average fate of seeds on *Banksia hookeriana* plants near Mt Adams for ages 0–25 years. Equations used in the model are described in the results and are listed in Table 1.

Gill & McMahon 1986; Cowling & Lamont 1987; Cowling *et al.* 1987; Bell *et al.* 1987). The annual production of cones and seeds increased rapidly between ages 5 and 15 years but then fluctuated around a mean of 17 cones per year. This early levelling-off of reproduction is in contrast to the continuous increase in seed production with age noted for *Banksia cuneata* to at least 23 years (Lamont *et al.* 1991b), and for *Banksia ornata* to as much as 38 years (Gill & McMahon 1986).

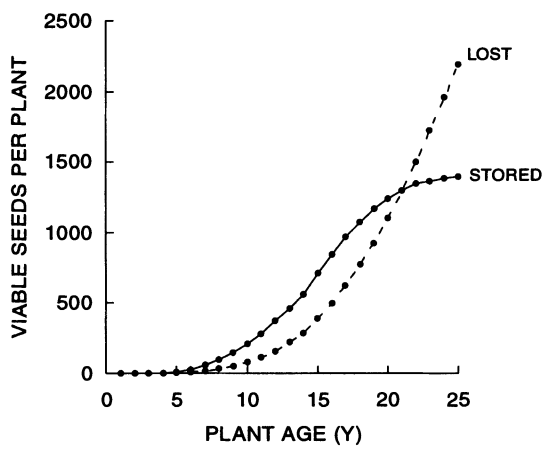


Fig. 4 The estimated total number of viable seeds stored in the crown (solid line), and the estimated total number of viable seeds lost to the combination of spontaneous follicle rupture, granivory and decay (broken line) in relation to plant age for *Banksia hookeriana* near Mt Adams, Western Australia (Calculated using the seed bank dynamics model described in the text and Fig. 3).

Despite large year to year variations in cone production, on average the total number of cones in the crown showed a steady increase with age. McMaster & Zedler (1981), Lamont *et al.* (1991a), and others hypothesized that canopy seed storage (serotiny) may dampen annual fluctuations in production, ensuring adequate seed supply following fire. The present results support this contention: seeds produced in years of high seed production were nearly all held on the plant for at least 5 years so that years of low seed production did not impact in the short term on the total canopy seed store.

Cowling *et al.* (1987) found that years of high seed production were associated with low rates of predispersal granivory by insects, implying satiation of granivores. If serotiny dampens fluctuations in seed availability, as suggested above, then these year to year variations would not prove effective in limiting the impacts of insect granivores unless granivory were

© 1996 British Ecological Society,

This content downloaded from 209.175.73.10 on Sat, 07 May 2016 01:46:40 UTC
All use subject to http://about.jstor.org/terms

restricted to follicles on cones less than one year old. Indeed, Scott (1982) claimed that granivory was concentrated on new cones since the largely host-specific insect larvae which eat *Banksia* seeds can only tunnel through the soft central axis of the inflorescence. He argued that the increasing woodiness of cones with age rendered those more than 1-year-old less vulnerable to invasion. However, Cowling *et al.* (1987) concluded that follicle damage from insects occurred in older cones for *Banksia menziesii* and *Banksia prionotes*. A similar finding was obtained here for *Banksia hookeriana* (Table 1): While granivory was greatest in 1-year-old follicles (18%), it continued at a rate of 2–3% of follicles per year in cones of increasing age. Zammit & Westoby (1988) also recorded granivory in old cones of *Banksia ericifolia* at Kuring-gai Chase National Park, New South Wales, but suggested that the trend might be explained by higher rates of infestation in the first few years of seed production when fewer inflorescences were available. However, levels of granivory in cones harvested from young stands provided only limited support for this explanation.

SEED BANK DYNAMICS AND FIRE FREQUENCY

According to Gill (1981) plants are not adapted to fire *per se*, but to a particular fire regime. The number of viable seeds stored in the crown of serotinous plants may largely determine the number offspring left after a fire. The balance between the accumulation of viable seeds in the canopy seed bank as plant age increases, and the loss of viable seeds to spontaneous follicle opening, granivory and decay might appropriately describe the optimum life history strategy for a non-sprouting species in relation to fire frequency. The canopy seed bank dynamics model estimates these patterns of accumulation and loss.

Early in the reproductive life of individuals of *Banksia hookeriana* (5–8 years of age) few viable seeds are produced, but few are lost. The absolute number of seeds available for dispersal following fire is small and there is a low likelihood of successful parent replacement given the high rates of seed and seedling mortality typical of the first few years following fire (Enright & Lamont 1989, 1992a,b). Using data on seedling recruitment after fire in Little Desert National Park, Victoria, Gill & McMahon (1986) estimated that parent replacement in *Banksia ornata* populations was unlikely until plants held ≈ 100 viable seeds in the crown. This correlated with a plant age of about 16 years. Independent evidence on recruitment of *Banksia hookeriana* near Mt Adams following experimental fires in 1983 and a wildfire in 1984 indicated that between 45 and 90 seeds were released for each recruit still alive in 1990 at the age of 6 or 7 years (Enright & Lamont 1992b). If it is assumed that 100 seeds are required to ensure parent replacement following fire, then the minimum postfire interval for *Banksia hookeriana* plants would be at least 8 years.

The total number of viable seeds lost from the canopy during the interfire period increased rapidly with age for plants > 10 years old. Nevertheless, the excess of total viable seeds produced in the plant's lifetime over total losses peaked at age 15 years. As plants continued to age in the absence of fire, total losses approached and finally exceeded the number of viable seeds still in the crown. This was predicted to occur for the average plant by the age of 22 years.

Using the simplifying assumption that self-replacement, on average, required release of at least 100 viable seeds per parent after fire, an estimate of the finite rate of natural increase (λ) can be calculated for each plant age (= fire frequency) with the maximum value of λ assumed to reflect the optimum strategy (λ as a measure of fitness, *sensu* Stearns 1992). Populations of *Banksia hookeriana* near Mt Adams appear optimally adapted to recurrent fires every 16–17 years (Fig. 5). Fires at this frequency also maximized the number of viable seeds released after fire relative to the total investment of energy and nutrients expended on seed production. Although the size of the canopy seed bank continued to grow until plants were about 25 years old, predicted λ decreased as plant age increased beyond 17 years. In addition, increasingly more resources are wasted on seeds which perish during the interfire period through spontaneous release into a highly competitive environment where successful recruitment is unlikely, and through pre-dispersal granivory and decay.

However, the predicted optimum fire frequency is dependent upon the estimated number of seeds needed to ensure parent replacement. If this estimate is reduced (greater likelihood of recruitment per seed) then the optimum occurs earlier, while if it is increased

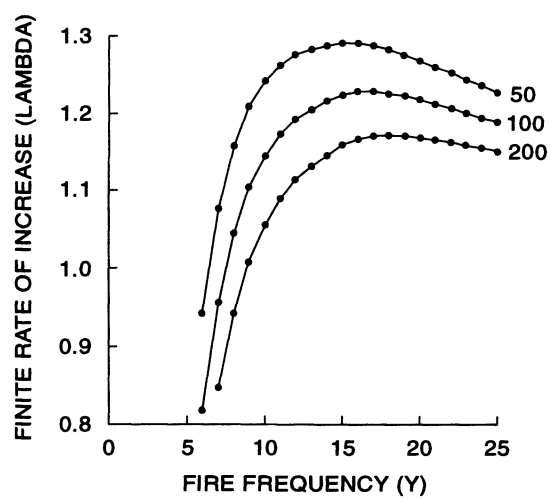


Fig. 5 Estimated finite rate of natural increase (λ) of *Banksia hookeriana* for different fire frequencies at Mt Adams, Western Australia assuming that 50, 100 or 200 viable seeds are required to ensure recruitment to reproductive age in the next generation.

This content downloaded from 209.175.73.10 on Sat, 07 May 2016 01:46:40 UTC
All use subject to http://about.jstor.org/terms

(less likelihood of recruitment) the optimum occurs later. It is likely that the true value varies from fire to fire, reflecting differences in the suitability of weather conditions for postfire recruitment (Enright & Lamont 1992a,b). For estimates ranging from half (50) to twice (200) the hypothesized number of seeds required per parent (relative to that used above), the optimum fire frequency ranges from 15 to 18 years, respectively (Fig. 5), and so appears quite robust.

If stands remain unburned for more than 25 years, opportunities for interfire recruitment might increase due to the release of sites within stands resulting from death of old individuals. However, Bond (1980) documents the decline of serotinous Proteaceae populations in South African fynbos which have remained unburned for 30–65 years, and we have recorded no successful interfire recruitment to reproductive age in *Banksia hookeriana* for stands up to 21 years old. The lack of older sites precludes further direct study of this question. The wastage of scarce resources (i.e. seed loss) in plants older than 22 years might imply that, historically, sites rarely reach this age before fire recurs. In support of this contention, we have been unable to locate any sites containing plants older than 23 years (Enright & Lamont unpublished data). However, this may also reflect a recent increase in fire frequency due to human interference.

Diversity and evenness in the canopy seed store modelled here had not peaked by the estimated optimum return time for fire. Nevertheless, diversity (H′) rises only slowly once this optimum is passed and evenness is always high. One possible consequence of dormant seed banks is the presence of seeds produced in different years which might reflect different pollen parents and selective forces (Harper 1977), i.e. it increases genetic diversity of the seed pool. This might help to buffer the species against rapid evolutionary drift, and at the same time increase resilience of the species in the face of fluctuating environmental conditions. These advantages apply to short-lived species characterized by soil seed storage where parent plants represent different generations. The advantages for longer lived, cohort regenerator species such as *Banksia hookeriana* are less obvious, as all or most pollen will derive from neighbours of identical age which were recruited following the same disturbance and have experienced similar growing conditions. Instead, selfing would probably increase with age as flower head production per plant and death of neighbours increased, cancelling out increasing seed age diversity. Cowling (1987) argued that serotinous nonsprouting species (such as *Banksia hookeriana*) are prone to strong selection pressure, particularly at the seedling stage. He considered that the high number of discrete sexual generations, and fragmentation of populations through fire-mediated local extinction, implies a rapid evolutionary rate and may explain the high level of speciation in *Banksia* and other taxa with similar life histories.

The recent history of fire within the geographical range of *Banksia hookeriana* suggests shorter intervals between successive fires than we propose as optimal for species' maintenance. A wildfire in January 1984 burned 117 000 ha of the northern sandplain heath (van der Moezel *et al.* 1987), an area equivalent to nearly half of the total geographical range of the species. In the autumn of 1991 wildfire again covered part of the 1984 burn area near Mt Adams, a return period of only 7 years. This fire eliminated *Banksia hookeriana* from one of the plots enumerated here. Van der Moezel *et al.* (1987) comment that areas of sandplain heath vegetation older than 11 years are becoming increasingly scarce since the establishment of farms and roads in the area and a hazard-reduction burning policy for the remaining natural vegetation.

Acknowledgements

This research was supported by Australian Research Council grants to BBL (1986–88) and BBL and NJE (1989–91). The authors acknowledge assistance in the field provided by Stephen van Leeuwen, Stephen Bergl and Ed Witkowski in particular, and the many students and research assistants from the School of Environmental Biology, Curtin University of Technology, Perth, who helped at various times. Insect species were identified by scientific staff of the Western Australian Museum, Perth.

References

Baker, H.G. (1989) Some aspects of the natural history of seed banks. *Ecology of Soil Seed Banks* (eds M. A. Leck, V. T. Parker & R. L. Simpson), pp. 9–23. Academic Press, San Diego.

Beard, J.S. (1984) Biogeography of the kwongan. *Kwongan: Plant Life of the Sandplain* (eds J. S. Pate & J. S. Beard), pp. 1–26. University of Western Australia Press, Nedlands.

Bell, D.T., Moezel, P.G. van der, Delfs, J.C. & Loneragan, W.A. (1987) Northern sandplain kwongan: effect of fire on *Hakea obliqua* and *Beaufortia elegans* population structure. *Journal of the Royal Society of Western Australia*, **69**, 139–143.

Bond, W.J. (1980) Fire survival of Cape Proteaceae – influence of fire season and seed predators. *Vegetatio*, **56**, 65–74.

Bradstock, R. (1985) *Plant population dynamics under varying fire regimes*. PhD thesis. University of Sydney, NSW.

Cowling, R.M. & Lamont, B.B. (1985a) Variation in serotiny of three Western Australian *Banksia* species along a climatic gradient. *Australian Journal of Ecology*, **10**, 345–350.

Cowling, R.M. & Lamont, B.B. (1985b) Seed release in *Banksia*: the role of wet-dry cycles. *Australian Journal of Ecology*, **10**, 169–171.

Cowling, R.M. & Lamont, B.B. (1987) Post-fire recruitment of four co-occurring *Banksia* species. *Journal of Applied Ecology*, **24**, 645–658.

Cowling, R.M. (1987) Fire and its role in coexistence and speciation in Gondwanan shrublands. *South African Journal of Science*, **83**, 106–112.

Cowling, R.M., Lamont, B.B. & Pierce, S.M. (1987) Seed

© 1996 British Ecological Society,

This content downloaded from 209.175.73.10 on Sat, 07 May 2016 01:46:40 UTC
All use subject to http://about.jstor.org/terms

bank dynamics of four co-occurring *Banksia* species. *Journal of Ecology*, **75**, 289–302.

Enright, N.J. & Lamont, B.B. (1989) Seed banks, fire season, safe sites and seedling recruitment in five co-occurring *Banksia* species. *Journal of Ecology*, **77**, 1111–1122.

Enright, N.J. & Lamont, B.B. (1992a) Survival, growth and water relations of *Banksia* seedlings on sand mine tailings and adjacent scrub-heath sites. *Journal of Applied Ecology*, **29**, 663–671.

Enright, N.J. & Lamont, B.B. (1992b) Recruitment variability in the resprouting shrub *Banksia attenuata* and non-sprouting congeners in the northern sandplain heaths of southwestern Australia. *Acta Oecologica*, **13**, 727–741.

George, A.S. (1981) The genus *Banksia*. *Nuytsia*, **3**, 239–473.

Gill, A.M. & McMahon, A. (1986) A post-fire chronosequence of cone, follicle and seed production in *Banksia ornata*. *Australian Journal of Botany*, **34**, 425–433.

Gill, A.M. (1981) Adaptive responses of Australian vascular plant species to fires. *Fire and the Australian Biota* (eds A. M. Gill, R. H. Groves & I. R. Noble), pp. 273–310. Australian Academy of Science, Canberra.

Harper, J.L. (1977) *Population Biology of Plants*. Academic, London.

Hnatiuk, R.J. & Hopkins, A.J.M. (1981) An ecological analysis of kwongan vegetation south of Eneabba, Western Australia. *Australian Journal of Ecology*, **6**, 423–438.

Lamont, B.B. & Barker, M.J. (1988) Seed bank dynamics of a serotinous, non-sprouting *Banksia* species. *Australian Journal of Botany*, **36**, 193–203.

Lamont, B.B. (1985) Fire responses of sclerophyll shrublands – a population ecology approach, with particular reference to the genus *Banksia*. *Ecology and Management of Fire in Natural Ecosystems of Western Australia* (ed. J. Ford), pp. 41–46. Environmental Studies Group, Western Australian Institute of Technology, Perth.

Lamont, B.B., Le Maitre, D.C., Cowling, R.M. & Enright, N.J. (1991a) Canopy seed storage in woody plants. *Botanical Review*, **57**, 277–317.

Lamont, B.B., Connell, S.W. & Bergl, S.M. (1991b) Seed bank and population dynamics of *Banksia cuneata*: the role of time, fire and moisture. *Botanical Gazette*, **152**, 114–122.

Lamont, B.B., Enright, N.J. & Bergl, S. (1989) Coexistence and competitive exclusion of *Banksia hookeriana* in the presence of congeneric seedlings along a topographic gradient. *Oikos*, **56**, 39–42.

McMaster, G.S. & Zedler, P.H. (1981) Delayed seed dispersal in *Pinus torreyana* (Torrey Pine). *Oecologia*, **51**, 62–66.

van der Moezel, P.G., Loneragan, W.A. & Bell, D.T. (1987) Northern sandplain kwongan: regeneration following fire, juvenile period and flowering phenology. *Journal of the Royal Society of Western Australia*, **69**, 123–132.

Scott, J.K. (1982) The impact of destructive insects on reproduction in six species of *Banksia*, L.f. (Proteaceae). *Australian Journal of Zoology*, **30**, 901–921.

Stearns, S.C. (1992) *The Evolution of Life Histories*. Oxford University Press, New York.

SYSTAT Inc. (1992) SYSTAT for Windows: Statistics, Version 5 edn. SYSTAT Inc., Evanston, IL.

Taylor, A. & Hopper, S.D. (1988) *The Banksia Atlas*. Australian Flora and Fauna Series Number 8, Australian Government Publishing Service, Canberra.

Witkowski, E.T.F., Lamont, B.B. & Connell, S.J. (1991) Seed bank dynamics of three co-occurring Banksias in south coastal Western Australia: The role of plant age, cockatoos, senescence and interfire establishment. *Australian Journal of Botany*, **39**, 385–397.

Witkowski, E.T.F., Lamont, B.B. & Obbens, F.J. (1994) Commercial picking of *Banksia hookeriana* in the wild reduces subsequent shoot, flower and seed production. *Journal of Applied Ecology*, **31**, 508–520.

Zammit, C. & Westoby, M. (1988) Pre-dispersal seed losses, and the survival of seeds and seedlings of two serotinous *Banksia* shrubs in burnt and unburnt heath. *Journal of Ecology*, **76**, 200–214.

Received 7 November 1994
revised version accepted 19 June 1995

© 1996 British Ecological Society,

This content downloaded from 209.175.73.10 on Sat, 07 May 2016 01:46:40 UTC
All use subject to http://about.jstor.org/terms